

Fig. 82.1 Reasons for ineffective health care provision during massive disasters. When damage to general and health care infrastructure combines with shortage of skilled health care workforce, deficient health care is inevitable. An increased number of patients further increases these drawbacks (list of included factors in each box is not exhaustive). (From Sever MS, Pawlowicz-Szklarska E, Tugular ZS. Nephrology in disasters, epidemics and wartime. In: Ronco C, Belomo R, House A, et al, eds. *ERA Neph-Mental*. European Renal Association; 2024. Accessed 3 March 2025. <https://e-learning.era-online.org/eboc/chapter-22-nephrology-in-disasters-epidemics-and-wartime/>.)

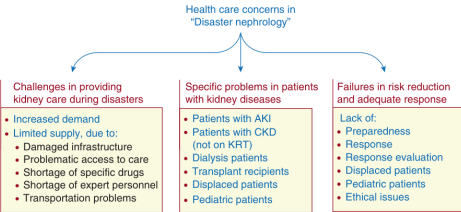
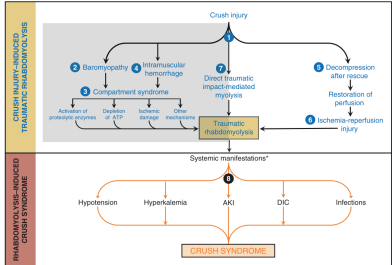


Fig. 82.2 Spectrum of health care problems in "disaster nephrology." During massive disasters, extra health care problems are added to routine demand, whereas health care provision is limited due to logistic reasons. Specific disaster-related problems in patients with kidney diseases exaggerate the dimensions of the crisis. Risk reduction by means of predisaster preparedness and response during the active phases of the disaster may only partially solve the problems. Retrospective evaluation of the response is mandatory to avoid similar mistakes in future disasters. Overall, drawbacks may be even more important in displaced patients and children. Of note, many ethical controversies exist in fair distribution of limited medical supplies and manpower. AKI, Acute kidney injury; CKD, chronic kidney disease; KRT, kidney replacement therapy.

[illegible]

Fig_82_1

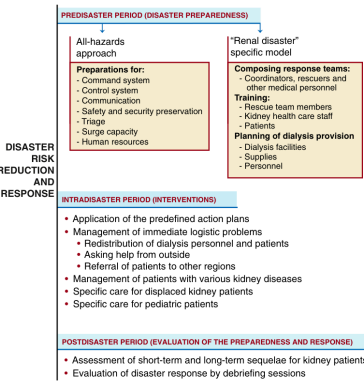


Fig. 82.4 An overview of risk reduction, response, and retrospective evaluation of the response in various phases of disasters. The all-hazards approach combined with the renal disaster-specific model in the predisaster period should be followed by application of predefined action plans and management of anticipated or instant/unexpected logistic problems. Management of patients with various problems related to kidney diseases, which are often even more severe in the postdisaster period for patients and in children, is anticipated and the avoidance of this stage in the predisaster period, overall predisaster preparations and intradisaster response are evaluated, shortcomings in response are identified, and measures are defined to avoid repetition in future disasters 37,39,109-141

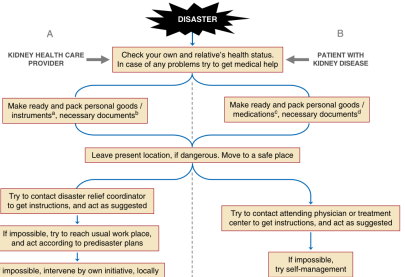


Fig. 8.2.5 Example of a simple, clear, and informative algorithm to minimize chaos and panic immediately after experiencing a destructive disaster (e.g., earthquake, war, hurricane). This algorithm describes early actions from the perspectives of kidney care providers (A) and patients with kidney diseases (B). Similar algorithms and/or infographics may (should) be created on various topics targeting the health care personnel or rescuers (e.g., interventions at the disaster field, issues to be considered for patient transport, indications for various medical interventions) or for the patients (e.g., how to secure themselves during the acute episode of a disaster, dietary suggestions, and simple instructions for self-treatment).

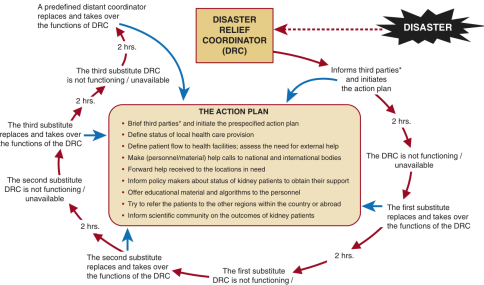
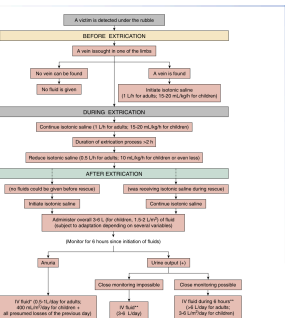


Fig. 62.6 **Rental disaster action plan emergency response and organizing relief.** Within the first two hours of the disaster, the disaster relief coordinator (DRC) should brief three parties to initiate the action plan. If the DRC cannot function or is not reachable within the first two hours, the first local substitute will be automatically assigned. If the first local substitute does not respond or cannot be reached, the second local substitute will act as the DRC. This scenario is repeated unless even the third local substitute cannot be reached. In the latter case, a predefined distant contact must be used to activate the disaster relief team. The disaster relief team consists of participants, particularly in the form of a communication process. Adaptations to the action plan considering the type and extent of the disaster, the status of the health care system, and the state of communications processes. Adaptations to the action plan considering the type and extent of the disaster and local circumstances may be necessary. (Adapted with permission from Sever MS, Lammie N, Vahrdt R. Rental disaster relief: from theory to practice. *Nephrol Dial Transplant*. 2009;24(8):1730–1735.)

Fig_82_4

[illegible]

Fig_82_7

Fig_82_5

Table B2.1 Differences in Various Features of Pediatric versus Adult Patients with Kidney Diseases During Disasters ^a	
Feature	Comments
Overall health concerns	- Malnutrition is more prominent in children - Children are more prone to communicable diseases in hygienic environments - Breakdowns in vaccination programs are more frequent in children - Prevalence of acute kidney more frequent
Medical concerns related with kidney care	- Perennial CKD usually more related - Incidence of crush syndrome-related AKI at lower in the first decade of life - Prevalence of pediatric patients on RRT is lower than adult patients - There is less well-defined patient care personnel - There is less compliance to dietary restrictions - Children are more susceptible to complications due to missed dietary restrictions - Children on pediatric-sized HD and PD equipment is more problematic - Availability of dialysis equipment formulations is more frequent - Shortage of specialized pediatric dialysis personnel is more prominent - Prevalence of pediatric patients are usually from chronic kidney disease plans - Many children are separated from their families and are unaccompanied
Logistic problems related to RRT patients	
General logistic and organizational problems	

AKI, acute kidney injury; HD, hemodialysis; RRT, kidney replacement therapy.

Table_82_1

Table 82.2 Tasks and Responsibilities of Relief Coordinators to Reduce the Risks for Patients with Kidney Diseases During Various Phases of Disasters^{a,b} 31,47,139,140

Before	• Create disaster intervention scenarios and action plans, and divide country into several sectors • Assign and train sector coordinators to intervene during disasters in their own sectors • Create financial support for disaster response • Create comprehensive algorithms to create medical and logistic problems • Create a protocol to conduct classification of relevant guidelines • Address how to include disaster medicine courses in medical and paramedical curricula and congresses • Create international organizations for collaboration in case of future disasters • Apply predefined measures to preserve own safety
During	• Implement preselected action plans (described in Fig. 12.8) immediately after the disaster • Apply algorithms to solve medical and logistic problems • Coordinate management of patients with various kidney diseases • Consider specific problems in children and disaster kidney patients • Follow ethical rules in all interventions
After	• Assess the efficiency of the disaster response and identify areas for future improvements in future disasters • Perform long-term inpatient and outpatient management in patients with kidney diseases to avoid similar mistakes in the future • Involve the scientific community on interventions and kidney patient outcomes

[†]This list is not exhaustive.

Table_82_2

Table 82.3 Tasks and Responsibilities of Rescuers and Kidney Health Care Providers to Reduce the Risks for Patients with Kidney Diseases Before, During, and After Disasters ^{a,b} 31,47,139,145,146	
Before	• Acquire and provide training on disaster nephrology • Disseminate relevant information to third parties • Train staff and organize drills for management of disaster-related problems • Train patients with kidney diseases on disaster preparedness and personal disaster response to minimize threats and drawbacks • Apply predefined measures to preserve own safety
During	• Apply prespecified interventions, but make adaptations in case of problems or inefficiency • Adapt treatment protocols for those injured or affected by the disaster in function of the logistic constraints • Refer patients to higher-level hospitals if local treatment is impossible • Evaluate local disaster response, specify shortcomings and make adaptations for improvement • Screen for and treat long-term disaster-related comorbidities • Inform the scientific community on interventions, and patient outcomes
After	
^a Data adapted from REF. ³¹ ^b This list is not exhaustive.	

Table_82_3

Table 82.4 Tasks and Responsibilities of Patients with Kidney Diseases to Reduce Risks Before, During, and After Disasters ^{a,b} 31,47,139,145,146	
Before	• Acquire education and training on how to cope with disaster-related medical conditions • Apply measures to decrease potential risks (e.g., maintaining a 2- week stock of medications and PD supplies) • Participate in drills on management of disaster-related problems
During	• Apply predefined measures to preserve own safety • Take predefined measures to minimize the health risks that are associated with disasters • Apply self-management protocols if attending physicians or nurses cannot be contacted
After	• Be vigilant for any changes in health status • Check status of own or alternative health care facilities for treatment possibilities
^a Data adapted from REF. ³¹ ^b This list is not exhaustive. <i>PD</i> , Peritoneal dialysis.	

Table_82_4

Table 82.5 Checklist for Information to Be Collected from Dialysis Units in the Context of Disaster Preparedness ³⁴	
• Name of the unit • Private/public status • Structural characteristics of the building • Possibility for helicopter landing within a perimeter of 1 km • Number and type of hemodialysis machines • Normal staffing • Number of nephrologists and other doctors and their contact information • Availability of pediatric nephrologists • Number of hemodialysis nurses and their contact information • Number of hemodialysis technicians and their contact information • Number of maintenance patients dialyzed in the unit; their name and contact information • Maximum number of patients, who can be treated in that particular unit • Availability of intensivists, intensive care positions, and possibilities to dialyze at the ICU	

Table_82_5

Table 82.6 “To Do List” for Nephrologists/Physicians for Treatment of Patients With Acute Kidney Injury During Various Phases of Disaster ^a	
Initial phase: intervention on the spot	• Detect, assess, and triage the victims, and perform a primary survey • Stop bleeding if any, resuscitate, and apply appropriate resuscitation if needed • Treat on-site life-threatening medical complications • Start fluid administration with consideration of the specific condition of the patient and disaster (Fig. 82.3) • Organize urgent transport to the nearest and most appropriate hospital, with appropriate measures when needed
At admission to the emergency unit of a reference hospital	• Register (if possible with the ICU, check vital signs, vital responses, triage) • Check type of fluids, and determine urgent electrolyte levels, if relevant, monitoring blood hypotension • Evaluate indications to admit to the ICU • After excluding urethral bleeding, insert a flexible catheter • Consult with relevant specialties • Make frequent visits • Avoid nephrotoxic medications • Evaluate evolution, and treat complications accordingly
Critical course: interventions during the critical course at the reference hospital	• Apply the most appropriate dialysis modality, considering medical and logistic features (Fig. 82.3) • Identify the necessity of unexpected findings or unexpected alterations in the clinical course
^a Although slight modifications may be required, overall, these principles go well for the entire spectrum of patients with kidney diseases and/or nondestructive disasters in a pandemic or not. Factors needed at an earlier stage (e.g., before medical intervention) are detailed in Fig. 82.3A. They include but are not limited to respiratory failure, hypotension, hypothermia, myocardial ischemia, and infection, all potentially fatal.	
^b Peritoneal dialysis relies on adequate contact of dialysate with the patient's blood, thereby requiring a certain degree of patient cooperation. In patients with severe trauma, inability to cooperate, or being unconscious, peritoneal dialysis is not recommended. In such cases, urgent transport to a hospital with appropriate facilities and equipment is recommended.	

Table_82_6

Table 82.7 Medical and Logistic Advantages and Drawbacks of Various Dialysis Modalities in Disaster Victims With Crush-Related Acute Kidney Injury ^{a,b,34,35}		
	Medical Advantages	Medical Drawbacks
IHD	• High clearance rate of low-molecular-weight solutes, like urea and potassium • Possibility for dialyzing without anticoagulation	• Priming volume may induce hypotension • High risk of disequilibrium syndrome
CRRT	• Better volume control • Low risk of disequilibrium syndrome • Possibility for higher calorie administration	• Need for continuous heparinization • Low removal capacity for small solutes, like potassium
PD	• No need for vascular access • Less hemodynamic instability	• Low clearance rate • Difficulty to maintain sterile conditions • Difficult application in some patients, especially those with severe thoracic and abdominal trauma
	Logistic Advantages	Logistic Drawbacks
IHD	• Possibility to treat several patients per day in the same position	• Need for experienced personnel and technical assistance
CRRT	• For CAVH application, no need for electricity and only minimal equipment is needed	• Treatment possibility for only one patient, per machine per day • Problems with transportation of fluid bags
PD	• For CAPD, no need for water and electricity	• Problems with transportation of fluid bags
^a CAPD, Continuous ambulatory peritoneal dialysis; CAVH, continuous arteriovenous hemodiafiltration; CRRT, continuous renal replacement therapy; IHD, intermittent hemodialysis; PD, peritoneal dialysis.		

Table_82_7